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Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Time: 03 Hours

Code: 100401 ***Subject: Biology for Engineers*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2x7=14]**

(a) Which of the following best describes the primary function of red blood cells in human body?
(i) To provide immunity
(ii) To help with digestion
(iii) To fight infections
(iv) To transport oxygen

(b) In recombinant DNA technology, which of the following enzymes is used to cut DNA at specific locations?
(i) Ligase
(ii) RNA polymerase
(iii) DNA polymerase
(iv) Restriction enzyme

(c) Which of the following is a primary concern regarding the use of genetically modified organisms (GMOs)?
(i) Ethical concerns and environmental impact
(ii) Reduced food production
(iii) higher levels of vitamins
(iv) Increased biodiversity

(d) Which of the following is used for comparing biological sequences, such as genes or proteins, to identify similarities or differences?
(i) Microarray analysis
(ii) PCR (Polymerase Chain Reaction)
(iii) Chromatography
(iv) BLAST (Basic Local Alignment Search Tool)

(e) In bioinformatics, which of the following databases would likely be used for storing protein structure information?
(i) Gen Bank
(ii) Protein Data Bank (PDB)
(iii) UniProt
(iv) Ensembl

(f) Which of the following materials is commonly used in medical implants due to its biocompatibility and corrosion resistance?
(i) Steel
(ii) Gold
(iii) Titanium
(iv) Aluminum

(g) A key property of biomaterials used in medical devices is:
(i) Biocompatibility
(ii) High reactivity
(iii) High conductivity
(iv) High porosity

(h) Bioremediation refers to the use of:
(i) Heat treatment for waste disposal
(ii) Chemical agents to neutralize toxins
(iii) Mechanical devices to extract toxins
(iv) Organisms to clean up environmental pollutants

(i) Which of the following is a major advantage of using algae in biofuels production?
(i) Low environmental impact
(ii) Easy cultivation on land
(iii) Absence of carbon dioxide emissions
(iv) High energy content

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(j) Which type of microorganisms is commonly used in the production of antibiotics?

(i) Bacteria (iii) Viruses

(ii) Protozoa (iv) Fungi

Q.2 Solve both questions : **[7+7=14]**

(a) What is bioremediation? Explain the principles of bioremediation in details.

(b) What is the importance of databases like Gen Bank in bioinformatics?

Q.3 Solve both questions : **[7+7=14]**

(a) Describe the four levels of protein structure (primary, secondary, tertiary, and quaternary).

(b) Explain the role of biosensors in detecting diseases. How can engineers use biosensors to detect pathogens or monitor health conditions?

Q.4 Solve both questions : **[7+7=14]**

(a) How can biofuels contribute to sustainable energy production? Explain in details.

(b) Explain the concept of protein folding. How do bioinformatics tools help predict the three-dimensional structure of proteins from their amino acid sequences?

Q.5 Solve both questions : **[7+7=14]**

(a) What is biofuel? Explain biofuels contribute to sustainable energy production.

(b) What are mutations in DNA? Discuss the various types of mutations (point mutation, insertion, deletion) and their potential effects of protein function.

Q.6 Solve both questions : **[4+10=14]**

(a) Define biomolecules and classify them into their major categories.

(b) What are the principles of bioremediation? Provide examples of how biotechnology is applied to clean up wastewater treatment.

Q.7 Solve this question : **[14]**

Discuss any exceptions to Mendel's laws of inheritance, including incomplete dominance, co-dominance, and epistasis. Explain with examples.

Q.8 Solve both questions : **[7+7=14]**

Discuss the role of microorganisms in biotechnology. Provide examples of how microorganisms are used in production of pharmaceuticals, biofuels, and food products.

Q.9 Write short notes on any two of the following:- **[7x2=14]**

(a) Explain Mendel's first law (Law of Segregation).

(b) What are macromolecules? Explain with examples.

(c) Explain the concept of enzyme inhibition.

(d) What is the role of cofactors and coenzymes in enzyme function?

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